Olympiad angle chasing

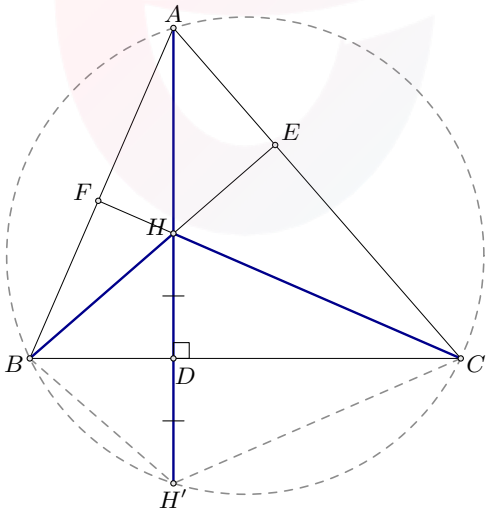
L4-GE1 Level 4 Olympiad · Geometry

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Cyclic quadrilaterals, directed angles and the standard configurations, used to write complete synthetic proofs.
By the end you can prove and use the converses of the cyclic and tangent criteria; run an angle chase with directed angles that works in every configuration; recognise orthocentre, antiparallel and incentre-excentre configurations and write BMO-standard proofs.
Prerequisites: Angle chasing and polygons (L2-GE1), Power of a point and cyclic quadrilaterals (L3-GE1). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

In an acute triangle ABC with orthocentre H, the reflection of H in BC lies on the circumcircle. To prove this, let BE and CF be altitudes. The right angles at E and F put A, F, H, E on a circle (the circle with diameter AH), so $\angle BHC = \angle FHE = 180^\circ - \angle A$. The reflection H' satisfies $\angle BH'C = \angle BHC = 180^\circ - \angle A$ and lies on the opposite side of BC from A, so the opposite angles of $ABH'C$ add to 180°, so $ABH'C$ is cyclic. This lemma is worth remembering, and we use the lemma again in Example 4.



The proof used right angles to get a circle, read an angle off that circle, and used the angle to get a second circle. Most olympiad angle chases alternate in this way between angle equalities and concyclicities. This unit covers the converses that make each step valid, directed angles to handle every configuration with one argument, the standard configurations, and what a marker expects in a written proof.

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1.1 Criteria and their converses

Cyclic criteria.
Four points A, B, C, D, no three collinear, are concyclic if and only if either of the following holds. Either $\angle ABD = \angle ACD$ with B, C on the same side of AD, or $\angle ABC + \angle ADC = 180^\circ$ with B, D on opposite sides of AC.
Tangent criterion. A line t through A touches circle ABC if and only if the angle between t and chord AB equals $\angle ACB$, the angle in the alternate segment.
Right angles. $\angle APB = 90^\circ$ if and only if P lies on the circle with diameter AB.

The forward directions are fairly trivial (the inscribed angle theorem). The method relies on the converses, and they follow from one observation. Fix the segment AD and a circle Γ through A and D. If P is inside Γ (on a given side of AD), extend AP to meet the arc at P'. Then $\angle APD$ is an exterior angle of triangle $PP'D$, so $\angle APD > \angle AP'D$. A point outside sees AD at a smaller angle in the same way. So on one side of AD, the points seeing AD at the angle given by the arc are exactly the points of that arc.

1.2 Directed angles

The weakness of these criteria is the side conditions. In a real problem a point may sit on a segment or beyond the segment, and the chase that says $\angle ABC + \angle ADC = 180^\circ$ in one picture says $\angle ABC = \angle ADC$ in another. We avoid this by measuring angles between lines, modulo 180°.

Directed angles.
For lines ℓ, m, let $\measuredangle(\ell, m)$ be the angle through which ℓ must be turned anticlockwise to become parallel to m, taken modulo 180°, and write $\measuredangle ABC = \measuredangle(BA, BC)$. Then $\measuredangle(\ell, m) = -\measuredangle(m, \ell)$, $\measuredangle(\ell, m) + \measuredangle(m, n) = \measuredangle(\ell, n)$, A, B, C are collinear iff $\measuredangle ABC = 0$, and $\measuredangle ABC + \measuredangle BCA + \measuredangle CAB = 0$ in every triangle.
Cyclic criterion. A, B, C, D are concyclic iff $\measuredangle ABC = \measuredangle ADC$. **Tangent criterion.** t touches circle ABC at A iff $\measuredangle(t, AB) = \measuredangle ACB$.

The directed cyclic criterion covers both cases. If B and D are on the same side of AC, triangles ABC and ADC have the same orientation, the directed angles are the ordinary ones with the same sign, and we recover equal angles. If they are on opposite sides, the orientations differ, say $\measuredangle ABC = \angle ABC$ and $\measuredangle ADC = -\angle ADC$, and equality modulo 180° means $\angle ABC + \angle ADC = 180^\circ$.

Two warnings. First, you cannot halve, because $2x = 2y$ only gives $x = y$ or $x = y + 90^\circ$, so central angles and angle bisectors need care. Route through a tangent instead. Second, directed angles forget which side of a line a point is on, so they cannot prove betweenness or compute "the angle is 40°". Use them for concyclicity, collinearity, parallels and tangency.

1.3 Standard configurations

The orthocentre. With feet D, E, F of the altitudes, the quadrilaterals $AFHE$, $BDHF$, $CEHD$ (diameters AH, BH, CH) and $BCEF$, $CAFD$, $ABDE$ (diameters BC, CA, AB) are all cyclic. Also $\angle BAH = 90^\circ - \angle B = \angle OAC$, so AH and AO are reflections of each other in the bisector of angle A.

Antiparallels. If X on line AB and Y on line AC make B, C, Y, X concyclic, then $\measuredangle(XY, AB) =$

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$\measuredangle YXB = \measuredangle YCB = \measuredangle ACB$, which is the angle the tangent at A makes with AB. So XY is parallel to that tangent, hence perpendicular to AO.

Incentre-excentre lemma.
Let line AI meet the circumcircle again at M, where I is the incentre and I_A the A-excentre. Then $MB = MC = MI = MI_A$.

Proof. M is the midpoint of arc BC, so $MB = MC$ and $\angle MBC = \angle MAC = \frac{1}{2}\angle A$. Hence $\angle MBI = \frac{1}{2}\angle A + \frac{1}{2}\angle B$, and $\angle MIB$, an exterior angle of triangle ABI, is also $\frac{1}{2}\angle A + \frac{1}{2}\angle B$. So $MB = MI$. The internal and external bisectors at B are perpendicular, so triangle IBI_A has a right angle at B, and the only point of the hypotenuse II_A equidistant from B and I is its midpoint, so M is that midpoint and $MI_A = MI$.

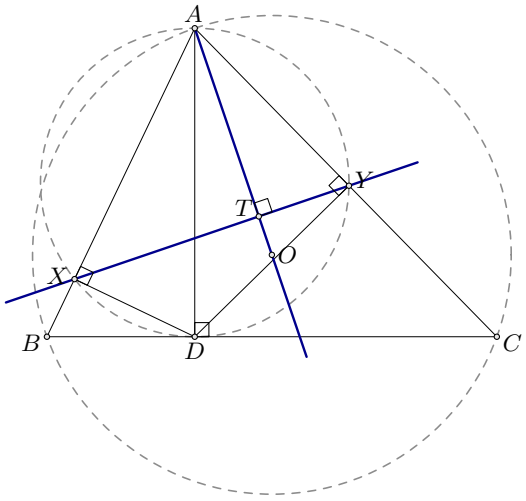
1.4 Worked examples

Example 1.
Let ABC be an acute triangle with circumcentre O, and let D be the foot of the altitude from A. Let X and Y be the feet of the perpendiculars from D to AB and AC. Prove that $XY \perp AO$.

Solution.
One could compute lengths such as $AX = AD^2/AB$, but an angle chase is shorter. Since $\angle AXD = \angle AYD = 90^\circ$, the points X and Y lie on the circle with diameter AD, so

$$\angle AYX = \angle ADX = 90^\circ - \angle DAB = \angle ABC.$$

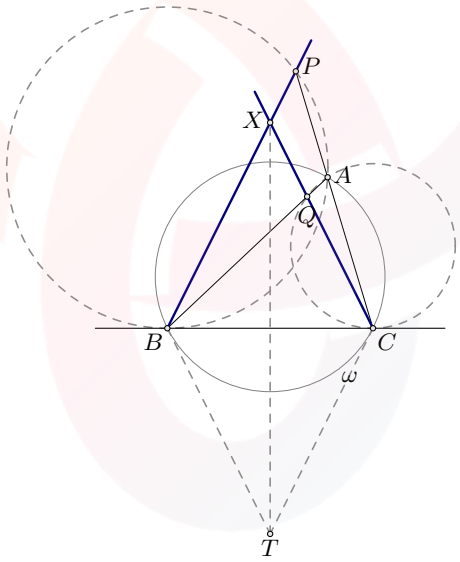
So B, C, Y, X are concyclic (exterior angle equals interior opposite angle), and by the antiparallel fact $XY \perp AO$. To finish by hand instead, note that triangle AOC is isosceles with apex angle $2\angle B$, so $\angle OAC = 90^\circ - \angle B$. If AO meets XY at T, triangle ATY has angles $90^\circ - \angle B$ at A and $\angle B$ at Y, leaving 90° at T.
Note that two right angles subtending a common segment give a circle on that segment as diameter (here the circle with diameter AD), and angles can then be transferred through that circle. You should look for such pairs before anything else.



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Example 2.
BC is a fixed chord of a circle ω, not a diameter, and A moves on ω. The circle through A tangent to line BC at B meets line AC again at P, and the circle through A tangent to BC at C meets line AB again at Q. Prove that lines BP and CQ pass through a fixed point.

Solution.
As A moves over both arcs, P can lie on segment AC, beyond A or beyond C, so an undirected chase would need cases. With directed angles one argument covers all of them.
By the tangent criterion in circle ABP, $\measuredangle(BC, BP) = \measuredangle BAP = \measuredangle BAC$, the last step because P lies on line AC. (Fixed angles for a moving point.) We look for an angle that does not change as A moves. By the cyclic criterion, $\measuredangle BAC$ is the same for every point A of ω, whichever arc A is on. Call this angle θ. So BP is the line through B at directed angle θ to BC, which is fixed. In the same way $\measuredangle(CB, CQ) = \measuredangle CAB = -\theta$, so CQ is also a fixed line.
Two fixed lines meet at a fixed point unless they are parallel, which needs $2\theta = 0$, that is $\angle BAC = 90^\circ$ and BC a diameter, which is excluded. The point can be identified. The tangent to ω at B makes angle $\measuredangle(BC, t_B) = -\theta$, so line BP is its mirror image in BC, and likewise for C. Hence the fixed point is the reflection in BC of the point T where the tangents at B and C meet.

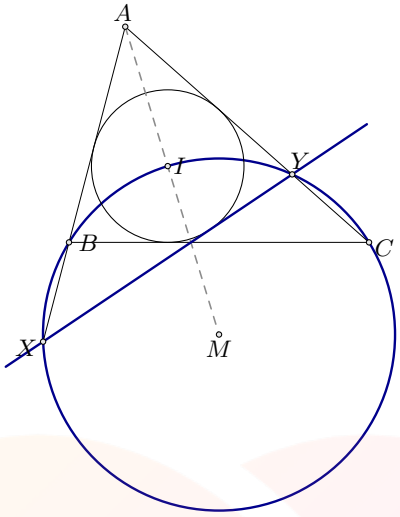


Example 3.
Let ABC be a triangle with $AB \neq AC$ and incentre I. The circle through B, I, C meets line AB again at X and line AC again at Y. Prove that XY is tangent to the incircle.

Solution.
Chasing gives $\angle AXY = \angle ACB$ quickly, since B, C, Y, X are concyclic, but tangency is a statement about the distance from I to XY, which angles at X and Y do not determine, and this is the hard step. Instead we use the incentre-excentre lemma, which says that circle BIC has centre M on line AI. A circle centred on a line is symmetric in that line, and reflection in AI swaps lines AB and AC.
So the reflection C^* of C in AI lies on the circle and on line AB, hence is B or X. The point C^* is not B, because $AC^* = AC \neq AB$. Thus $X = C^*$ and similarly $Y = B^*$, so line XY is the mirror image of line BC. The incircle is its own mirror image and touches BC, so the incircle touches XY. The argument also shows $AX = AC$ and $AY = AB$.

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(Reflection in an axis of symmetry.) When a circle's centre lies on an axis of symmetry of the figure, reflect.



Example 4.
Let ABC be an acute triangle with circumcentre O. The circle through B, O, C meets lines AB and AC again at D and E, different from B and C. Prove that the reflection of O in DE lies on the circle through A, D, E.

Solution.
The conclusion resembles the fact proved at the start of the unit, so we guess that O is the orthocentre of ADE, and aim for $DO \perp AE$. Depending on the shape of ABC, D may lie on segment AB or beyond B, so we use directed angles.
Computing $\measuredangle OCB$ is the hard step. The familiar value $90^\circ - \angle A$ comes from halving $\angle BOC = 2\angle A$, which directed angles do not allow. Use the tangent t at C instead. Since $t \perp CO$,

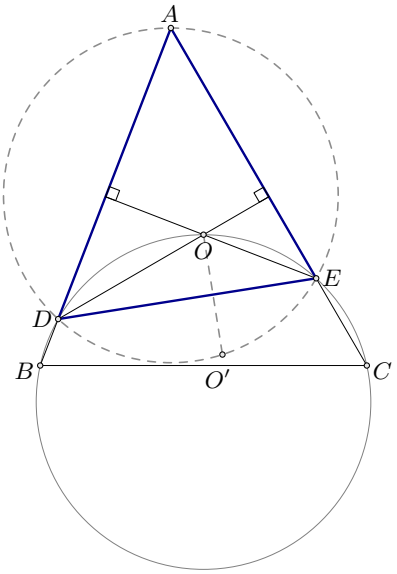
$$\measuredangle(CO, CB) = \measuredangle(CO, t) + \measuredangle(t, CB) = 90^\circ + \measuredangle CAB.$$

Now circle $BOCD$ and the line ABD give

$$\measuredangle(DO, AC) = \measuredangle ODB + \measuredangle(AB, AC) = \measuredangle OCB + \measuredangle BAC = 90^\circ + \measuredangle CAB + \measuredangle BAC = 90^\circ.$$

So $DO \perp AE$, and swapping the roles of B and C gives $EO \perp AD$, so O is the orthocentre of ADE. (The triangle is non-degenerate, since A cannot lie on circle BOC, as A sees BC at $\angle A$, not $2\angle A$.) To finish without asking whether ADE is acute, redo the opening proof with directed angles. Turning both lines through 90° gives $\measuredangle DOE = \measuredangle(AE, AD) = \measuredangle EAD$. Reflection reverses directed angles, so the reflection O' satisfies $\measuredangle DO'E = \measuredangle DAE$, and A, D, E, O' are concyclic.

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1.5 Writing up the solution

A BMO marker wants a proof that is correct in every configuration and checkable line by line. Say at the start if you use directed angles modulo 180°. Give a reason for every equality (which circle, which collinearity, which tangent). Prove a concyclicity before you use the circle. Deal with degenerate cases in a sentence. A diagram is a guide, never a justification.

Common traps.

- Proving the result only for your drawing when a point could lie on a segment or its extension.
- Halving directed angles, for instance with central angles or bisectors.
- Using "the points look cyclic" in place of a criterion.
- Circular reasoning. When the claim is that P lies on a circle, do not draw P on the circle and read off angles. Prove an angle equality, or define P' on the circle and show $P' = P$.
- Trying to get betweenness or an actual angle size out of directed angles.

1.6 How to spot this in a contest

Use these methods when the conclusion is concyclicity, collinearity, parallel or perpendicular lines, or tangency, and the data are angles, right angles, tangents or circles. I would work in the following order.

1. Mark every right angle and every tangent, since each gives a circle or an angle equality.
2. If you see an orthocentre, an incentre, an arc midpoint or a circumcentre, recall the standard facts above before chasing.
3. If the chase stalls, draw an accurate figure, look for four points that seem concyclic but have not been joined yet, and prove that as a lemma (an auxiliary circle).
4. If a point moves, ask which directed angle stays fixed.

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2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆⋆ to ⋆⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆⋆ Let ABC be a triangle and let D be a point on side BC, other than B and C. The tangent at D to the circle ABD meets line AC at E, and the tangent at D to the circle ACD meets line AB at F. Prove that $EF \parallel BC$.

Problem 2 ⋆⋆⋆ Let ABC be an acute triangle with orthocentre H, and let E and F be the feet of the altitudes from B and C. A line ℓ through H, other than AH, meets the circle ABH again at X and the circle ACH again at Y (take $X = H$ or $Y = H$ if ℓ touches that circle). Prove that the midpoint of XY lies on the circle AEF.

Problem 3 ⋆⋆⋆⋆ Let ABC be a triangle with no right angle and circumcentre O, and let D be a point on side BC, other than B and C. The perpendicular bisector of BD meets line AB at P, and the perpendicular bisector of CD meets line AC at Q. Prove that A, P, O, Q are concyclic.

Problem 4 ⋆⋆⋆⋆ Let ABC be a triangle with incentre I and circumcircle ω, and let line AI meet ω again at M. Let P be a point of ω other than A, B, C, M, and let line PM meet line BC at X. Prove that the circle through P, X, I is tangent to line AI.

Problem 5 ⋆⋆⋆⋆⋆ Let ABC be a triangle with $AB \neq AC$ and incentre I. The incircle touches BC at D, and M is the midpoint of BC. Let K and L be the feet of the perpendiculars from B to line CI and from C to line BI. Prove that D, K, L, M are concyclic.

Problem 6 ⋆⋆⋆⋆⋆ Let ABC be a triangle, and let W be a point on line BC other than B and C. The circle through W tangent to line AB at B and the circle through W tangent to line AC at C meet again at X. Prove that, as W varies, line WX passes through a fixed point.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **BMO1 2013/14 Problem 2** ⋆⋆
   Two right angles on a common segment give a circle, then bring in the tangent.
2. **BMO1 2015/16 Problem 2** ⋆⋆
   Alternate segment theorem, then an exterior angle of the cyclic quadrilateral.
3. **BMO1 2010/11 Problem 3** ⋆⋆
   Show that two angles at L add up to 180 degrees, reading each off a circle.
4. **AIME I 2003 Problem 10** ⋆⋆
   Angle chasing alone does not finish, so add an auxiliary point that creates an isosceles or equilateral triangle.
5. **BMO1 2019/20 Problem 3** ⋆⋆
   Draw the common tangent at the point of contact and chase through the tangent.

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6. **BMO1 2023/24 Problem 3** ⋆⋆⋆
   Isosceles triangles at B plus inscribed angles, and directed angles avoid case splits.

7. **AIME I 2016 Problem 6** ⋆⋆⋆
   Incentre-excentre lemma, then similar triangles.

8. **BMO1 2014/15 Problem 5** ⋆⋆⋆
   Arc midpoints give equal angles, so look for a cyclic quadrilateral through P and Q.

9. **BMO1 2015/16 Problem 5** ⋆⋆⋆
   Many right angles mean many circles, so prove collinearity by angles.

10. **USAJMO 2011 Problem 5** ⋆⋆⋆
    Bring in the centre of the circle and find a circle through P and the tangency points.

11. **USAJMO 2016 Problem 1** ⋆⋆⋆⋆
    Guess the fixed point from a special position of P, and arc midpoints are centres of circles through incentres.

12. **USAJMO 2018 Problem 3** ⋆⋆⋆⋆
    Reflections make B equidistant from three points, so check whether Q lies on a line through B.

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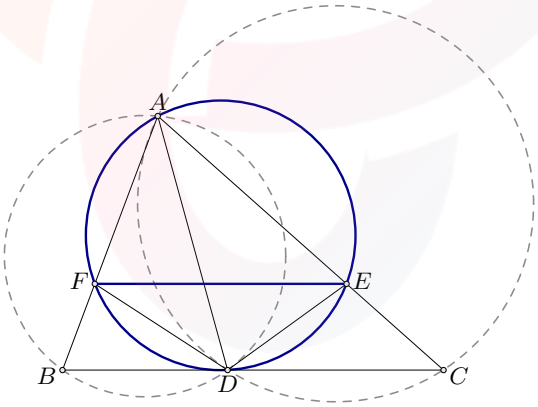
4 Hints

1. First show that A, E, D, F are concyclic, by computing $\angle EDF$ from the two tangents.
2. Compute the directed angles that lines XA and YA make with ℓ, using $BH \perp AC$ and $CH \perp AB$.
3. Draw the circle with centre P through B and D, and the circle with centre Q through C and D. Show that their second common point lies on the circumcircle.
4. First prove $MB^2 = MX \cdot MP$ by showing that MB touches the circle BXP. Then bring in the incentre-excentre lemma.
5. Find a circle through B, D, I, K and show that BC bisects angle KDL. Then use $MK = ML$.
6. Show that X lies on the circumcircle. Then look at $\angle BXW$, which does not depend on W.

5 Solutions

Problem 1

Let ABC be a triangle and let D be a point on side BC, other than B and C. The tangent at D to the circle ABD meets line AC at E, and the tangent at D to the circle ACD meets line AB at F. Prove that $EF \parallel BC$.



We use directed angles modulo $180°$, since E and F may lie on the sides or on their extensions. Note that $E \neq A$, because the tangent at D meets circle ABD only at D, and similarly $F \neq A$.

The tangent criterion in circle ABD and in circle ACD gives

$$\angle(DE, BC) = \angle DAB, \qquad \angle(DF, BC) = \angle DAC.$$

Subtracting, $\angle EDF = \angle DAB - \angle DAC = \angle(AC, AB) = \angle EAF$, the last step because E lies on AC and F on AB. Hence A, E, D, F are concyclic.

Now read the angle at F off this circle.

$$\angle(FE, AB) = \angle EFA = \angle EDA = \angle(DE, BC) + \angle(BC, AD) = \angle(AD, AB) + \angle(BC, AD) = \angle(BC, AB).$$

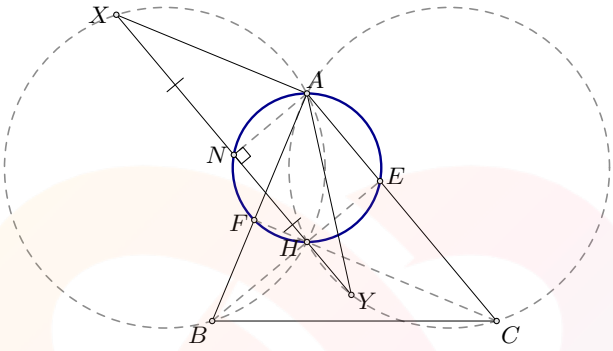
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So $EF \parallel BC$. ■

Remark. Two tangents at the same point D give two angles at D, and their difference is an angle of a quadrilateral through A. The circle $AEDF$ is not in the statement, and finding the circle is the main step.

Problem 2

Let ABC be an acute triangle with orthocentre H, and let E and F be the feet of the altitudes from B and C. A line ℓ through H, other than AH, meets the circle ABH again at X and the circle ACH again at Y (take $X = H$ or $Y = H$ if ℓ touches that circle). Prove that the midpoint of XY lies on the circle AEF.



We use directed angles modulo $180°$, since X and Y may lie on either side of H. As $\ell \neq AH$, neither X nor Y is A. Since X lies on circle ABH and $BH \perp AC$,

$$\measuredangle(XA, \ell) = \measuredangle AXH = \measuredangle ABH = \measuredangle(AB, AC) + \measuredangle(AC, BH) = \measuredangle(AB, AC) + 90°,$$

with the tangent criterion in place of the cyclic criterion if $X = H$. In the same way, as $CH \perp AB$,

$$\measuredangle(YA, \ell) = \measuredangle ACH = \measuredangle(AC, AB) + 90°.$$

So $\measuredangle(XA, \ell) = -\measuredangle(YA, \ell)$. Let N be the foot of the perpendicular from A to ℓ, and reflect the figure in line AN. The reflection maps ℓ to itself and reverses directed angles, so the image of line AX is the line through A at directed angle $-\measuredangle(XA, \ell)$ to ℓ, which is line AY. Hence the reflection maps X to Y, and N is the midpoint of XY.

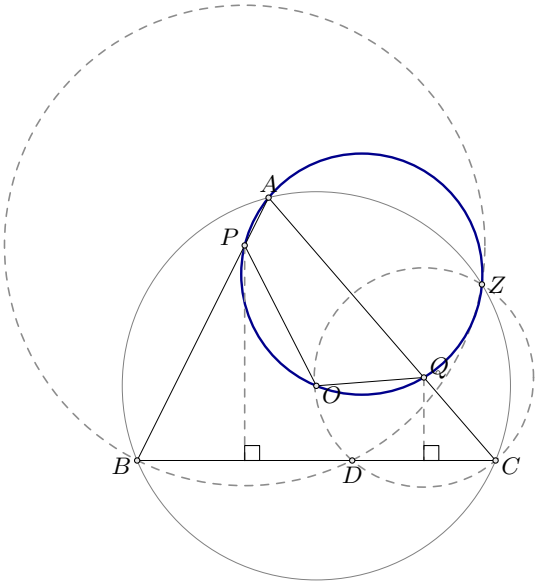
Finally $\angle ANH = 90°$, or $N = H$, so N lies on the circle with diameter AH. The right angles at E and F put E and F on the same circle, which is therefore the circle AEF. ■

Remark. The reflections of H in AB and AC lie on the circumcircle, so circles ABH and ACH are the reflections of the circumcircle in AB and AC and have its radius R. The sine rule in each gives $AX = 2R \sin \angle AHX = 2R \sin \angle AHY = AY$, which is the same isosceles triangle seen through lengths.

Problem 3

Let ABC be a triangle with no right angle and circumcentre O, and let D be a point on side BC, other than B and C. The perpendicular bisector of BD meets line AB at P, and the perpendicular bisector of CD meets line AC at Q. Prove that A, P, O, Q are concyclic.

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We use directed angles modulo 180°. Let γ_P be the circle with centre P through B and D, and γ_Q the circle with centre Q through C and D. The tangent t_B to γ_P at B is perpendicular to the radius PB, that is to AB, and likewise the tangent t_C to γ_Q at C is perpendicular to AC.

Step 1. The circles meet again at a point Z other than B, C, D. Line DP is the reflection of line BP in the perpendicular bisector of BD, so $\measuredangle(DP, BC) = \measuredangle(BC, AB)$, and in the same way $\measuredangle(DQ, BC) = \measuredangle(BC, AC)$. If the circles touched at D, then P, D, Q would be collinear and these two angles would be equal, which gives $AB \parallel AC$. So there is a second common point $Z \neq D$. Also $Z \neq B$, since otherwise Q would be equidistant from the three collinear points B, D, C, and similarly $Z \neq C$.

Step 2. Z lies on the circumcircle. The tangent criterion in γ_P and in γ_Q gives

$$\measuredangle BZD = \measuredangle(t_B, BC) = 90^\circ + \measuredangle(AB, BC), \qquad \measuredangle DZC = \measuredangle(BC, t_C) = \measuredangle(BC, AC) + 90^\circ.$$

Adding, $\measuredangle BZC = \measuredangle(AB, AC) = \measuredangle BAC$, so Z lies on circle ABC and $OZ = OB = OC$.

Step 3. The points O and P are both equidistant from B and Z, so if $P \neq O$ the line OP is the perpendicular bisector of BZ. In the same way $OQ \perp CZ$ if $Q \neq O$. Turning both lines through 90°,

$$\measuredangle POQ = \measuredangle(ZB, ZC) = \measuredangle BAC = \measuredangle PAQ,$$

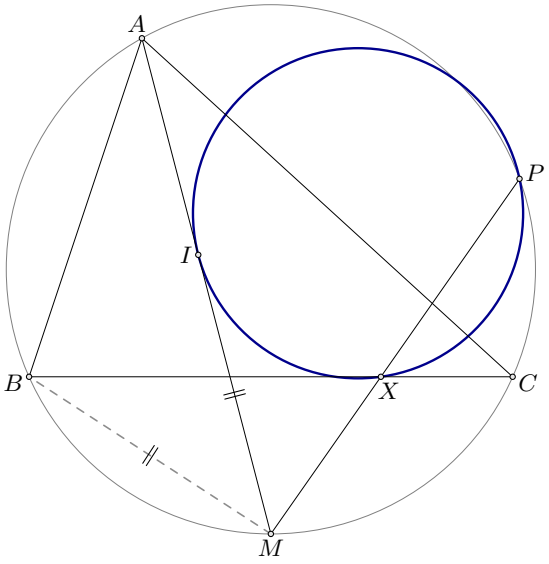
as P lies on AB and Q on AC. So A, P, O, Q are concyclic. If P or Q coincides with A or O, fewer than four distinct points remain and there is nothing to prove. ■

Remark. A point on the perpendicular bisector of BD is the centre of a circle through B and D, so draw that circle (an auxiliary circle). Once Z is on the circumcircle, all three of O, P, Q are centres of circles through Z, and their connecting lines are perpendicular to the common chords.

Problem 4

Let ABC be a triangle with incentre I and circumcircle ω, and let line AI meet ω again at M. Let P be a point of ω other than A, B, C, M, and let line PM meet line BC at X. Prove that the circle through P, X, I is tangent to line AI.

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We use directed angles modulo 180°, since P may lie on either arc BC. As M is the midpoint of arc BC, the tangent to ω at M is parallel to BC, so line PM is not parallel to BC and X exists. Also $X \neq B$ (otherwise $P = B$), $X \neq P$ (as P is not on BC), and I is not on line PM, since line MI is line MA and $P \neq A$.

Step 1. Since $MB = MC$, triangle MBC is isosceles, so $\measuredangle MBC = \measuredangle BCM = \measuredangle BPM$, the last step in ω. As X lies on BC and on PM, this reads $\measuredangle(BM, BX) = \measuredangle BPX$, so by the tangent criterion MB touches the circle BXP at B. Hence M lies outside this circle, X and P are on the same ray from M, and power of a point gives

$$MB^2 = MX \cdot MP.$$

Step 2. By the incentre-excentre lemma $MI = MB$, so $MI/MX = MP/MI$. The triangles MIX and MPI share the angle at M, so they are similar, with opposite orientations. Hence

$$\measuredangle(AI, IX) = \measuredangle MIX = \measuredangle IPM = \measuredangle IPX,$$

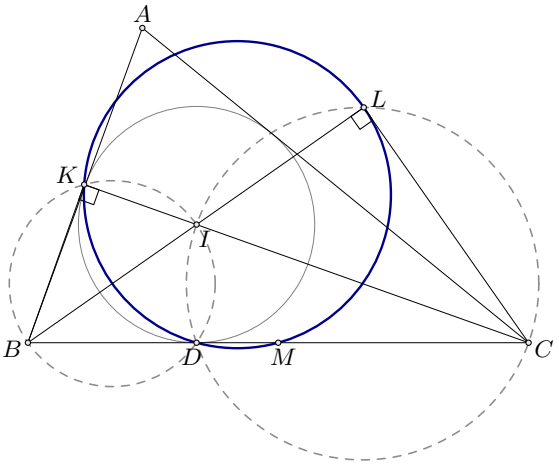
and by the tangent criterion line AI touches the circle PXI at I. ■

Remark. Step 1 is the shooting lemma, $MB^2 = MX \cdot MP$ for every chord through the arc midpoint M. The incentre-excentre lemma then moves the tangent from B to I, and the same proof works with the excentre I_A in place of I.

Problem 5

Let ABC be a triangle with $AB \neq AC$ and incentre I. The incircle touches BC at D, and M is the midpoint of BC. Let K and L be the feet of the perpendiculars from B to line CI and from C to line BI. Prove that D, K, L, M are concyclic.

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We use directed angles modulo 180°. Since $\angle BIC = 90^\circ + \frac{1}{2}\angle A$ is not a right angle, $K \neq I$ and $L \neq I$, and $K \neq L$. As $\angle BCI = \frac{1}{2}\angle C$ is acute, K lies on ray CI and differs from C, so K is on the same side of BC as A, and the same holds for L.

Step 1. BC bisects angle KDL. The right angles $\angle BDI = \angle BKI = 90^\circ$ put D and K on the circle with diameter BI, so

$$\measuredangle(DK, BC) = \measuredangle KDB = \measuredangle KIB = \measuredangle(CI, BI).$$

In the same way D and L lie on the circle with diameter CI, and $\measuredangle(DL, BC) = \measuredangle LDC = \measuredangle LIC = \measuredangle(BI, CI)$. So $\measuredangle(DK, BC) = -\measuredangle(DL, BC)$, and lines DK and DL are reflections of each other in BC. These lines are different, because a line equal to its own reflection is BC or perpendicular to BC, while $\measuredangle(CI, BI)$ is neither 0 nor 90°. Let Γ be the circle DKL.

Step 2. A phantom point. Let line BC meet Γ again at M', with $M' = D$ if BC touches Γ. Then

$$\measuredangle KLM' = \measuredangle KDM' = \measuredangle(DK, BC) = \measuredangle(BC, DL) = \measuredangle M'DL = \measuredangle M'KL,$$

using the tangent criterion for the first and last steps when $M' = D$. In triangle KLM' these are the angles at L and K, both measured in the sense of the triangle's orientation, so they are equal as ordinary angles and $M'K = M'L$.

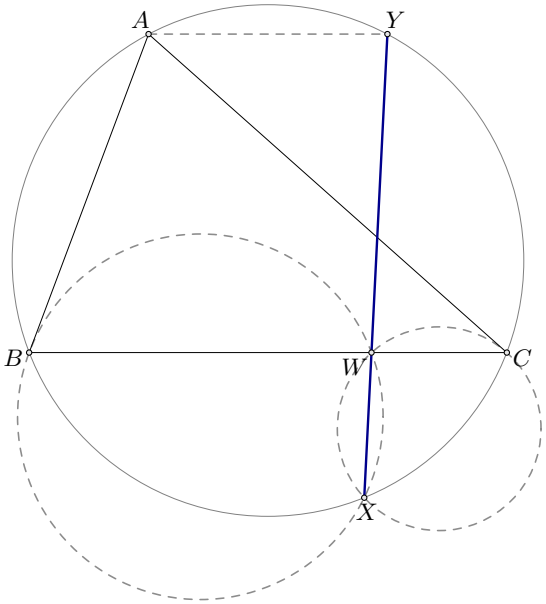
Step 3. The right angles $\angle BKC = \angle BLC = 90^\circ$ put K and L on the circle with diameter BC, so $MK = ML$. Thus M and M' both lie on line BC and on the perpendicular bisector of KL. If these two lines did not meet in a single point, KL would be a chord perpendicular to the diameter BC, and BC would bisect the chord, putting K and L on opposite sides of BC. So $M = M'$, and M lies on Γ. ■

Remark. Step 2 is the lemma that the bisector of an angle of a triangle and the perpendicular bisector of the opposite side meet on the circumcircle. The hypothesis $AB \neq AC$ only keeps $M \neq D$.

Problem 6

Let ABC be a triangle, and let W be a point on line BC other than B and C. The circle through W tangent to line AB at B and the circle through W tangent to line AC at C meet again at X. Prove that, as W varies, line WX passes through a fixed point.

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We use directed angles modulo 180°, since W may lie on segment BC or outside, and X then moves between the two arcs BC. Let γ_B and γ_C be the two circles and ω the circumcircle of ABC.

First, X exists and differs from W, B and C. The tangents at the two ends of a chord are reflections of each other in the perpendicular bisector of the chord. So if γ_B and γ_C touched at W with common tangent t, the chord BW of γ_B would give $\measuredangle(t, BC) = -\measuredangle(AB, BC)$, the chord CW of γ_C would give $\measuredangle(t, BC) = -\measuredangle(AC, BC)$, and $AB \parallel AC$ would follow. Also line BC meets γ_C only at C and W, so $B \notin \gamma_C$ and $X \neq B$, and similarly $X \neq C$.

Step 1. X lies on ω. The tangent criterion in γ_B and in γ_C gives

$$\measuredangle BXW = \measuredangle(AB, BC), \qquad \measuredangle WXC = \measuredangle(BC, AC).$$

Adding, $\measuredangle BXC = \measuredangle(AB, AC) = \measuredangle BAC$, so X lies on ω.

Step 2. The fixed point. Let Y be the point where the line through A parallel to BC meets ω again, with $Y = A$ if this line touches ω (that is, if $AB = AC$). Note $Y \neq B$. If $X = Y$ there is nothing to prove. Otherwise the cyclic criterion in ω, or the tangent criterion at A when $Y = A$, gives

$$\measuredangle BXY = \measuredangle BAY = \measuredangle(AB, BC),$$

since $AY \parallel BC$. Comparing with Step 1, $\measuredangle BXY = \measuredangle BXW$, so X, Y, W are collinear. Hence every line WX passes through Y, the reflection of A in the perpendicular bisector of BC. ■

Remark. A point X that sees BC at a fixed directed angle runs over a fixed circle, and if line XW makes a fixed directed angle with XB, the line passes through a fixed point of that circle. To name the point, read the fixed angle at A instead of at X.